

Review

Assessing the Potential of Artificial Intelligence in Advancing Clean Energy Technologies in Europe: A Systematic Review

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Abstract: This systematic review investigates the role of artificial intelligence (AI) in advancing clean energy technologies within Europe, based on a literature survey from 2006 to 2023. The assessment reveals that AI, particularly through deep learning and neural networks, enhances the efficiency, optimization, and management of clean energy systems. Noteworthy is AI's capacity to improve short-term energy forecasts, essential for smart cities and IoT applications. Our findings indicate that AI drives innovation in renewable energy, contributing to the development of smart grids and enabling collaborative energy-sharing models. While the research underscores AI's substantial influence in Europe's energy sector, it also identifies gaps, such as varied AI algorithm applications in different renewable energy sectors. The study emphasizes the need for integrating AI with emerging clean energy innovations, advocating for interdisciplinary research to navigate the socio-economic, environmental, and policy dimensions. This approach is crucial for guiding a sustainable and balanced advancement in the clean energy landscape, signifying AI's pivotal role in Europe's energy transition.

Keywords: artificial intelligence; clean energy technologies; systematic review; European context; renewable energy; machine learning



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1. Introduction

The escalating concerns about climate change and the urgency for sustainable energy solutions have intensified global efforts towards clean energy technologies. Europe stands at the forefront of this movement, setting ambitious targets to curtail greenhouse gas emissions and pivot towards a sustainable energy framework. Meeting these targets necessitates the integration of cutting-edge technologies and innovations that bolster the evolution and adoption of clean energy solutions. Among these, artificial intelligence (AI) has surfaced as a potent instrument [1], poised to expedite the uptake of clean energy technologies [2] and play a pivotal role in Europe's energy metamorphosis.

AI's breadth, encompassing machine learning and deep learning, has seen substantial advancements recently, fueling its application across various sectors with a notable impact on the energy sector. In the clean energy domain, AI is expected to enhance efficiency and facilitate informed decision-making. Russell et al. (2016) [1] and Vinuesa et al. (2020) [2] discuss AI's potential to drive sustainable energy goals. AI's significant role in future carbon-neutral energy systems is becoming increasingly clear, where it aids in designing and discovering novel materials for clean energy applications, as explored by Tabor et al. (2018) [3]. In the realm of clean energy, AI promises enhancements ranging from heightened efficiency [4] to strategic resource distribution [5] and informed decision-making [6,7]. Specifically, AI's role as a linchpin in future carbon-neutral energy systems is undeniable [8]. A salient application of AI in this context is its capability to assist in the design and discovery of novel materials tailored for clean energy applications [9]. Yet, the journey of integrating AI into clean energy is not devoid of challenges, encompassing issues like data security, regulatory intricacies, and the volatile nature of weather patterns [4,10].

Recent studies have explored various facets of sustainable development and technology. For instance, Wang et al. delved into the impact of geopolitics on international scientific cooperation, focusing on US–China marine pollution research [11], and the influence of geopolitics on environmental research [12]. Another study by Wang and Huang assessed the repercussions of the COVID-19 pandemic on sustainable development goals [13]. The integration of blockchain technology into the energy sector has also been explored, emphasizing its potential in revolutionizing energy systems [14]. Furthermore, the interplay between the digital economy and carbon dioxide emissions has been examined, shedding light on the role of threshold variables in this relationship [15].

While the literature underscores the vast potential of AI in clean energy, it also highlights discernible research gaps. A conspicuous void is the dearth of exhaustive studies focusing on AI-assisted material discovery for clean energy. Although AI-driven techniques are acknowledged for their cost-effectiveness and speed in material design for clean energy, a comprehensive understanding of AI's future role in this sector remains elusive [9]. Our research endeavors to fill this void, presenting a unique perspective through a comprehensive analysis of AI-assisted material discovery. We delve into the nuances of AI's potential in material design for clean energy, offering a fresh lens to comprehend its future trajectory.

This study is committed to bridging the identified gaps, presenting a meticulous review of the current research landscape on AI's potential in propelling clean energy technologies in Europe. Through an exhaustive examination of research papers and conference proceedings, this review aims to illuminate the most promising AI applications in clean energy. It will also elucidate the benefits, challenges, and prospects tied to their deployment. By identifying research clusters, discerning trends, and tracing thematic evolutions, this review aspires to be an invaluable resource for researchers, policymakers, and industry stakeholders steering Europe's clean energy transition.

2. Review of the Scientific Literature

2.1. AI in the Development and Deployment of Clean Energy Technologies

Recent advancements underscore the transformative potential of Artificial Intelligence (AI) in the realm of clean energy technologies. AI not only enhances the efficiency and productivity of these technologies through real-time data analysis, automation, and intelligent decision-making, but also plays a pivotal role in material discovery and public engagement.

In the context of Europe, one of the gaps identified in the literature is the intersection of gender equality and clean energy policies in the European Union (EU). There is a need for more research to mainstream gender within the EU clean energy transition [16].

Lastly, there is a gap in the literature on the implications of the use of AI in smart electricity systems from the perspective of energy justice. More research is needed to explore how legal norms or the existing literature can provide a framework to apply this concept to AI technologies [17].

Maleki et al. (2022) [4] delve into how AI-assisted materials design and discovery can introduce new efficient materials for clean energies, emphasizing the acceleration of the materials discovery process. This acceleration leads to the development of better-performing and cost-effective clean energy technologies. Tabor et al. (2018) [3] propose an integrated AI approach towards autonomous materials discovery. Their approach, which is expected to reduce the time and cost associated with materials development, leverages AI to explore the vast materials design space more efficiently than conventional methods. This enables the discovery of novel materials with properties that are highly desirable for clean energy applications.

In the realm of energy management, Ahmad et al. (2021) [7] delve into the application of AI for managing solar and hydrogen power generation, emphasizing AI's superior performance over traditional models. They highlight AI's advantages in areas such as enhancing controllability, managing large datasets, bolstering cybersecurity, optimizing smart grids, and advancing IoT and robotics. Furthermore, AI's contribution to energy

efficiency and predictive maintenance underscores its potential to revolutionize clean energy systems.

AI's capabilities extend to adjusting energy production based on dynamic variables such as weather patterns and demand forecasts, which is crucial for the optimization of smart grids. Studies like those by Xu et al. (2021) [18] and Zhang et al. (2022) [19] provide insight into the scientific foundations and bibliometric analyses of AI in renewable energy, reflecting its growing influence. Similarly, the works of Sulaiman et al. (2023) [20] explore the security aspects of AI applications in power grids, indicating a move towards more resilient energy infrastructures.

The potential for AI to maximize energy output and minimize waste is demonstrated by Liu et al. (2022) [21], who discuss the challenges and future perspectives of integrating renewables into multi-energy systems for a carbon-neutral transition. Meanwhile, Ekinici et al. (2022) [22] offer practical insights through their experimental investigation into solar PV panel maintenance, a crucial aspect of photovoltaic efficiency.

Beyond the direct management of energy systems, AI's influence permeates to adjacent sectors that indirectly impact energy efficiency. For instance, Lee et al. (2022) [23] and Li et al. (2022) [24] examine the role of industrial robots in promoting green technology innovation, while Li et al. (2021) [25] analyze the structural characteristics and determinants of international green technological collaboration networks. These studies collectively reinforce the integral role of AI in shaping a more sustainable and efficient future for energy management.

Furthermore, AI's role in public engagement is evident in the work of Buah et al. (2020) [6], who introduce a communication and engagement framework for energy projects that utilizes AI. This framework aids project managers in developing virtual agents for engaging citizens and stakeholders, showcasing how AI can be instrumental in promoting public involvement in clean energy projects, ensuring their successful implementation, and enhancing public acceptance.

AI can also improve the performance of solar panels by detecting issues such as shading, dust, or debris accumulation on solar arrays, ensuring panels operate at their maximum capacity and increasing overall energy yield [21]. AI-powered grid management, with its advanced algorithms and machine learning techniques, processes vast volumes of real-time data, enabling grid operators to monitor energy consumption patterns, track renewable energy generation, and assess grid performance in real-time [26]. Such capabilities of AI open up new avenues for intelligent energy management systems, aiding in waste reduction, efficient energy storage, and accelerating the green energy transition [24]. By harnessing the power of AI, both businesses and individuals can optimize their energy usage, reduce wastage, and make strides towards a sustainable future [22].

2.2. Benefits of AI in Clean Energy Technologies

AI offers several benefits in clean energy technology, as evidenced by the studies discussed earlier. Maleki et al. (2022) [4] highlight the role of AI-assisted materials design and discovery in developing new efficient materials for clean energies. Ahmad et al. (2021) [7] emphasize the use of AI in solar and hydrogen power generation, supply and demand management control. Energy efficiency optimization in smart buildings is discussed by [27]. Chen et al. (2021) [28] propose an AI-based useful evaluation model for forecasting renewable energy and energy efficiency impact on the economy, demonstrating AI's potential in improving decision-making and planning processes in the energy sector.

According to Mazzeo et al. (2021) [29], the utilization of Artificial Neural Networks (ANNs) can be suggested for sizing and simulating a clean energy community (CEC), which includes a PV–wind hybrid system, energy storage systems, and electric vehicle charging stations to meet the building district energy demand. This method highlights the potential of AI to enhance the efficiency and environmental impact of clean energy systems.

2.3. Challenges to AI Implementation in Clean Energy Technologies

While AI has the potential to aid in the development of clean energy technologies, it also presents challenges that must be addressed. Maleki et al. (2022) [4] discuss the advantages and challenges of using AI in material discovery for clean energy. Ahmad et al. (2021) [7] and Asif (2020) [30] further stress the importance of considering the unpredictability of weather patterns in the implementation of AI-based clean energy technologies.

Several scholarly articles delve into the challenges of implementing AI in clean energy technologies. “AI in Energy: Overcoming Unforeseen Obstacles” [31] offers a multidisciplinary approach, emphasizing the lack of regulation and standards for AI in energy systems and the challenge of scalability, especially when considering safety and environmental protection standards. “Artificial intelligence in sustainable energy industry: Status Quo, challenges and opportunities” [7] provides a baseline for understanding AI’s current status in the sustainable energy industry, touching upon challenges like data availability, algorithm selection, and the ethical and social implications. Another study, “Artificial intelligence powered large-scale renewable integrations in multi-energy systems for carbon neutrality transition: Challenges and future perspectives” [21], focuses on the practical applications of AI in renewable energy systems, underscoring the challenges of accurate forecasting and the need for effective energy storage solutions. Lastly, “Artificial intelligence-based solutions for climate change: a review” [32] examines how AI can enhance energy efficiency, discussing its potential in predicting energy demand, optimizing energy production, and fostering sustainable development. Collectively, these articles underscore the multifaceted challenges of AI in clean energy, from data accuracy and algorithm selection to regulatory compliance and ethical considerations.

2.4. Opportunities for AI in Clean Energy Technology

There are significant opportunities for using AI in clean energy technology. Maleki et al. (2022) [4] and Ahmad et al. (2021) [7] both emphasize how AI-assisted materials’ design and discovery can help develop new materials for clean energy applications. Li (2021) [33] discusses the potential of an AI-powered Energy Internet to accelerate carbon neutrality and transform the energy industry. AI can play a key role in developing sustainable clean energy solutions. By enhancing the efficiency and performance of clean energy technologies, optimizing resource allocation, and facilitating better decision-making, AI has the potential to significantly contribute to Europe’s clean energy transition. Continued research and investment in AI applications for clean energy technologies will be crucial for overcoming the challenges and maximizing the opportunities presented by AI in the pursuit of a more sustainable and resilient energy system.

Several scholarly articles highlight the transformative potential of Artificial Intelligence (AI) in the clean energy sector. “Artificial intelligence in sustainable energy industry: Status Quo, challenges and opportunities” [7] underscores AI as a pivotal tool in the evolving energy industry, optimizing systems, cutting costs, and enhancing efficiency. The global impact of the COVID-19 pandemic on energy dynamics and the intertwined opportunities with AI technologies are explored in “Crises and opportunities in terms of energy and AI technologies during the COVID-19 pandemic” [34]. “AI in Energy: Overcoming Unforeseen Obstacles” [31] emphasizes the necessity of a coordinated approach, serving as a guide for policymakers, energy enterprises, and scholars. Delving into the broader implications, “How artificial intelligence will affect the future of energy and climate” [35] contemplates AI’s influence on energy supply–demand and its ramifications on climate change. Lastly, “AI: the secret to unlocking the potential of renewable energy?” [36] envisions AI as a cornerstone in the renewable energy sector’s future, aiding in cost reduction, efficiency maximization, and the shift from fossil fuels. Collectively, these works project AI as a game-changer in the energy domain, with its promise overshadowed only by the challenges of regulation and standardization, necessitating collaborative efforts for a sustainable energy trajectory.

3. Research Methodology

The systematic review methodology was chosen for its rigorous approach to identifying, evaluating, and synthesizing all relevant studies on a particular topic. This approach is particularly suitable for our research question, which seeks to assess the potential of AI in advancing clean energy technologies in Europe. Previous studies, such as those by Wang et al. (2023) [11] and Wang and Huang (2021) [13], have successfully employed systematic reviews to provide comprehensive insights into their respective fields.

3.1. Literature Search

A comprehensive literature search was conducted across three leading academic databases: Scopus, Clarivate, and IEEE Xplore. Scopus offers broad interdisciplinary coverage, vital for examining Europe-centric research, while Clarivate's Web of Science provides a curated collection of high-impact papers, ensuring the quality of the sourced literature. IEEE Xplore contributes focused technological insights, particularly in AI advancements. The search phrases were formulated to capture the essence of AI's role in clean energy within the European context, spanning articles published between 2006 and 2023.

The literature search on the role of artificial intelligence in clean energy development and deployment within the context of Europe was conducted across three leading academic databases: Scopus, Clarivate, and IEEE Xplore. These databases were selected due to their comprehensive coverage of the scientific literature and their reputations as reliable sources for academic research in various fields, including artificial intelligence, clean energy, and European policy. The search phrases included combinations of keywords such as "artificial intelligence", "machine learning", "clean energy", "renewable energy", "sustainable energy", "Europe", "European Union", "European countries", and "European nations". The search (Table 1) was limited to articles published between 2006 and 2023.

Table 1. Search results overview: database and keywords used in literature selection.

Database/Keywords Used to Search	Results	Results
Scopus	AND "Europe"	
"Artificial intelligence" AND "clean energy"	1	147
"Machine learning" AND "renewable energy"	18	1890
"Intelligent control" AND "sustainable energy"	1	30
"Deep learning" AND "green energy"	0	70
"Data analytics" AND "renewable resources"	0	10
Clarivate:		
TS = ("Artificial intelligence") AND TS = ("clean energy")	0	65
TS = ("Machine learning") AND TS = ("renewable energy")	16	1240
TS = ("Intelligent control") AND TS = ("sustainable energy")	0	19
TS = ("Deep learning") AND TS = ("green energy")	0	35
TS = ("Data analytics") AND TS = ("renewable resources")	0	3
IEEE Xplore:		
("Artificial intelligence" OR "Machine learning" OR "Deep learning") AND ("Clean energy" OR "Renewable energy" OR "Green energy")	121	3650
("Intelligent control" OR "Data analytics") AND ("Sustainable energy" OR "Renewable resources")	2	133

3.2. Article Selection

Articles were screened for relevance based on their titles and abstracts. The inclusion and exclusion criteria were established to ensure that the selected articles directly addressed the research objectives. The manual screening process ensured a thorough and unbiased selection of articles. The Critical Appraisal Skills Programme (CASP) criteria were employed to assess the relevance and quality of the articles, a method that has been validated and used in previous research to ensure the rigor and quality of systematic reviews.

The search results were screened for relevance by examining the titles and abstracts of the identified articles. Articles were included in the review if they met the following criteria: (a) focused on the application of AI in clean energy technologies, (b) addressed the European context, and (c) were written in English. Duplicates, conference abstracts, and non-peer-reviewed articles were excluded. After the initial screening, the full texts of the selected articles were examined to ensure their relevance and to extract the necessary data.

After conducting a thorough evaluation using the Critical Appraisal Skills Programme (CASP) criteria, we meticulously assessed the relevance and quality of the 244 articles initially identified for this study. This rigorous appraisal process led to the exclusion of 39 articles. The primary reason for their exclusion was that these articles addressed subjects that were tangentially related or not directly aligned with the core research objectives of our study. As a result, they were deemed not pertinent to the specific focus of our systematic review. Consequently, we retained a total of 205 articles that were more closely aligned with our research purposes and provided a comprehensive insight into the application of artificial intelligence in advancing clean energy technologies in Europe. These selected articles formed the foundation for our subsequent analysis and synthesis.

The initial screening and selection process was conducted manually to ensure a comprehensive and relevant set of articles. However, the subsequent data extraction and thematic analysis employed the use of Bibliometrix, an R package, to facilitate a systematic examination of the collected literature.

3.3. Data Extraction

Data extraction was guided by a standardized form, ensuring consistency and comprehensiveness in the gathered information. The variables selected for extraction, such as AI techniques and clean energy technologies, were chosen based on their relevance to the research question and their prominence in the existing literature. For instance, AI techniques like deep learning have been highlighted in prior studies, such as Wang and Su (2020) [14], for their potential in the energy sector.

Data were extracted from the selected articles using a standardized data extraction form. The extracted data included the publication year, author names, article title, research objectives, AI techniques used, clean energy technologies studied, European countries addressed, findings, and any identified challenges or limitations. The extracted data were compiled and organized in a structured format to facilitate the subsequent thematic analysis.

3.4. Thematic Analysis and Quality Assessment

Thematic analysis was employed to identify and synthesize recurring themes in the literature. This approach was chosen for its ability to provide a nuanced understanding of complex datasets. The combination of deductive and inductive coding allowed for a comprehensive exploration of both anticipated and emergent themes. The use of Bibliometrix, an R package, was inspired by its successful application in prior bibliometric studies, such as Wang et al.'s exploration of geopolitics in environmental research [12].

The quality of the articles was continuously assessed using the CASP checklist, a tool that has been widely recognized and employed in systematic reviews for its robustness in evaluating research quality.

The extracted data were subjected to a thematic analysis to identify recurring themes, patterns, and trends in the literature. The analysis was performed using a combination of deductive and inductive approaches. Deductive coding was based on predefined categories, such as AI techniques, clean energy technologies, and European countries, while inductive coding was used to identify emerging themes and patterns not covered by the predefined categories. The coding process was iterative, with new codes being added and existing codes being refined as needed. The final set of codes was used to group the findings into distinct themes, which were then used to synthesize the overall findings of the review.

Bibliometrix, an R package for quantitative research in scientometrics and bibliometrics, was employed during the analysis process. This tool facilitated the systematic

examination of the collected literature, allowing for the identification of research trends, topic clusters, and thematic evolutions in the field.

Throughout the research process, the quality of the included articles was assessed using the Critical Appraisal Skills Programme (CASP) checklist, which evaluates aspects such as the research question, study design, data collection, data analysis, and conclusions. The results of the quality assessment informed the interpretation and synthesis of the findings.

4. Results and Discussion

Table 2 provides a detailed overview of the research landscape from 2006 to 2023, capturing the breadth and depth of academic contributions in the domain of AI's application in clean energy within Europe. The metrics presented in this table serve to contextualize the volume, diversity, and collaborative nature of research in this field. By understanding the distribution of document types, the extent of collaboration among authors, and the prevalent keywords, we aim to paint a comprehensive picture of the research dynamics over the years.

Table 2. Overview of research landscape: key metrics and document characteristics (2006–2023).

Description	Results	Description	Results
MAIN INFORMATION ABOUT DATA		AUTHORS	
Timespan	2006:2023	Authors	2130
Sources (Journals, Books, etc.)	185	Authors of single-authored docs	21
Documents	244	AUTHOR COLLABORATIONS	
Annual Growth Rate %	21.65	Single-authored docs	26
Document Average Age	3.25	Co-authors per doc	9.11
Average citations per doc	17.34	International co-authorships %	30.33
References	11,574	DOCUMENT TYPES	
DOCUMENT CONTENTS		Article	121
Keywords Plus (ID)	2208	Conference paper	93
Author's Keywords (DE)	804	Conference review	5
		Review	25

For instance, the high number of articles compared to reviews indicates a vibrant and active research community producing primary research. The average citations per document and the annual growth rate further underscore the increasing importance and recognition of this research area. The international co-authorship percentage highlights the collaborative and cross-border nature of research in this domain, emphasizing the global interest and efforts in AI and clean energy.

In essence, this table serves as a backdrop, setting the stage for the subsequent detailed analysis, by offering a macroscopic view of the research trends, collaborations, and focal points in the domain of AI and clean energy in Europe.

Table 3 presents predefined categories derived from the terms frequently encountered in the literature on artificial intelligence's role in clean energy. These categories serve as a foundation for a deductive approach in thematic analysis, helping to structure and organize the vast array of topics covered in the research. By categorizing these terms, we can gain a clearer understanding of the primary areas of focus and the interrelationships between various concepts in the realm of AI and clean energy.

Table 3. Predefined categories for deductive analysis of AI in clean energy literature.

Category	Terms	Frequency
AI Techniques	artificial intelligence, deep learning, machine learning, learning systems, long short-term memory, learning algorithms, machine learning, neural networks, reinforcement learning, artificial neural network	79, 63, 32, 38, 22, 13, 13, 13, 10, 9
Energy Types	clean energy, wind power, renewable energies, energy utilization, green energy, renewable energy resources, solar energy, renewable energy source, solar power generation, alternative energy	41, 32, 29, 28, 21, 21, 17, 14, 15, 9
Energy Management and Efficiency	energy efficiency, energy management, energy management systems, energy conservation, energy storage	24, 20, 12, 11, 11
Technology and Infrastructure	internet of things, smart power grids, big data, smart grid, data analytics, deep neural networks, data mining, smart city	24, 20, 14, 12, 11, 11, 9, 10
Policy and Economics	energy policy, sustainable development, energy, economics, investments	21, 23, 14, 12, 12
Applications and Use Cases	forecasting, electric power transmission networks, decision making, optimization, weather forecasting, decision support systems	50, 29, 17, 16, 10, 9
Energy Sources and Technologies	fossil fuels, electric power generation, photovoltaic cells	15, 10, 9
Environmental Impact	climate change	12
Other	electric utilities, energy resources	9, 9

The categories encompass a range of topics from specific AI techniques to different energy types, management strategies, and broader societal implications. Such a structured approach aids in identifying overarching themes, potential gaps, and emerging trends in the literature, ensuring a comprehensive review of the subject matter.

4.1. Trend Topics

In the rapidly evolving landscape of energy and artificial intelligence (AI), it is crucial to understand the trends and shifts in academic and industry discussions. By analyzing the frequency and temporal distribution of specific topics in the literature, we can gain insights into the areas that have garnered attention over the years and predict future trajectories. The analysis depicted by Figure 1 delves into various topics related to energy and AI, shedding light on their prominence and relevance over time.

Figure 1 provides a comprehensive overview of the selected topics, detailing their frequency of occurrence and distribution across years. Each topic's first, median, and third quartiles offer a snapshot of its prominence over time, allowing us to discern patterns and shifts in the discourse.

1. Electric Power Distribution: This topic was most prominent around 2014, with a shift towards more recent discussions by 2018. The spread from the first quartile to the third quartile suggests a resurgence of interest in the later years.
2. Intelligent Control: This topic had earlier prominence around 2009, peaking around 2015, and then seeing a decline by 2016.
3. Wind Turbines: Discussions around wind turbines have been consistent from 2014, peaking in 2016, and extending until 2019.
4. Electric Vehicles: This topic has seen a broad span of interest from as early as 2008, with a median in 2016, and discussions extending into 2021.
5. MATLAB: The use or discussion of MATLAB in the context of energy and AI peaked around 2015–2018.
6. DC–DC Converters: This topic has gained traction more recently, from 2017 to 2021.

7. Energy Conversion: Similar to DC–DC converters, this topic has been more prevalent in recent years, from 2018 to 2021.
8. Renewable Resource: Discussions peaked around 2016–2018 and extended to 2020.
9. Energy Policy: This topic has seen significant interest, especially from 2017 to 2021.
10. Artificial Intelligence: As expected, AI has seen a surge in discussions, especially from 2017 onwards, peaking in 2022.
11. Learning Systems, Wind Power, Forecasting, Clean Energy, Renewable Energies, Deep Learning, Machine Learning, and Green Energy: All these topics have seen a significant rise in discussions, especially post-2019, indicating a growing interest in the intersection of AI and energy in recent years.
12. Data Mining: This topic has a more recent focus, peaking in 2023.

Topics like artificial intelligence, deep learning, and machine learning have seen a significant surge in recent years, indicating the growing integration of these technologies in the energy sector. Clean energy, renewable energies, and green energy are also trending topics, reflecting the global shift towards sustainable energy solutions. Traditional topics like electric power distribution and intelligent control have seen varied interest over the years but remain relevant. The focus on energy policy suggests that as technology evolves, there is a parallel discussion on the regulatory and policy aspects of these advancements. Overall, the data reflect the evolving landscape of energy and AI, with a clear trend towards sustainable energy solutions and the integration of advanced AI technologies in this domain.

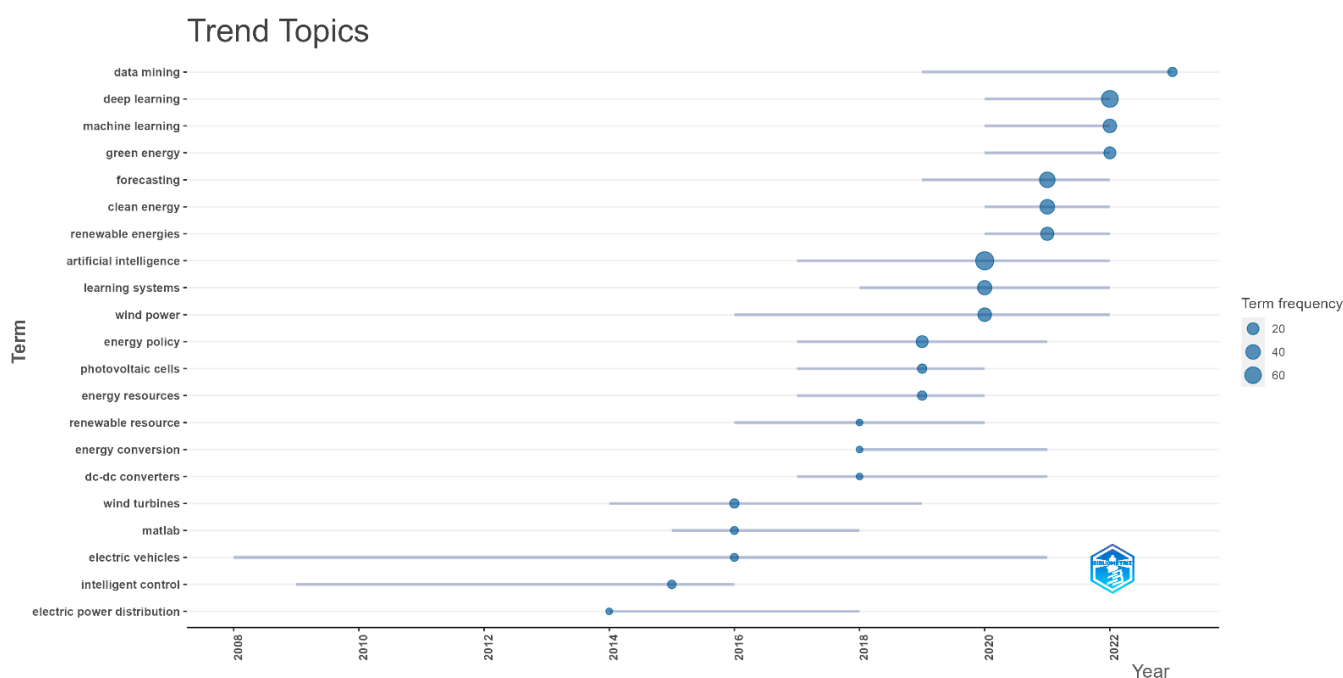


Figure 1. Temporal trends of key topics in energy and artificial intelligence (2008–2023).

4.2. Historiography

The articles are centered around sustainable energy, artificial intelligence, and their applications in various domains like solar energy prediction, battery management, and nuclear reactors. We employed a historiography (Figure 2) analysis, which uses the Global Citation Score (GCS) that represents the total number of citations received by a particular document or set of documents in the dataset. It is a straightforward metric that provides a cumulative count of how often a work has been referenced by other works. The Global Citation Score (GCS) offers a metric of scholarly impact, reflecting the degree to which specific articles resonate within the broader academic community. For instance, the arti-

cle titled “Accelerating the Discovery of Materials for Clean Energy in the Era of Smart Automation” has a GCS of 371, indicating a high level of recognition and citation.

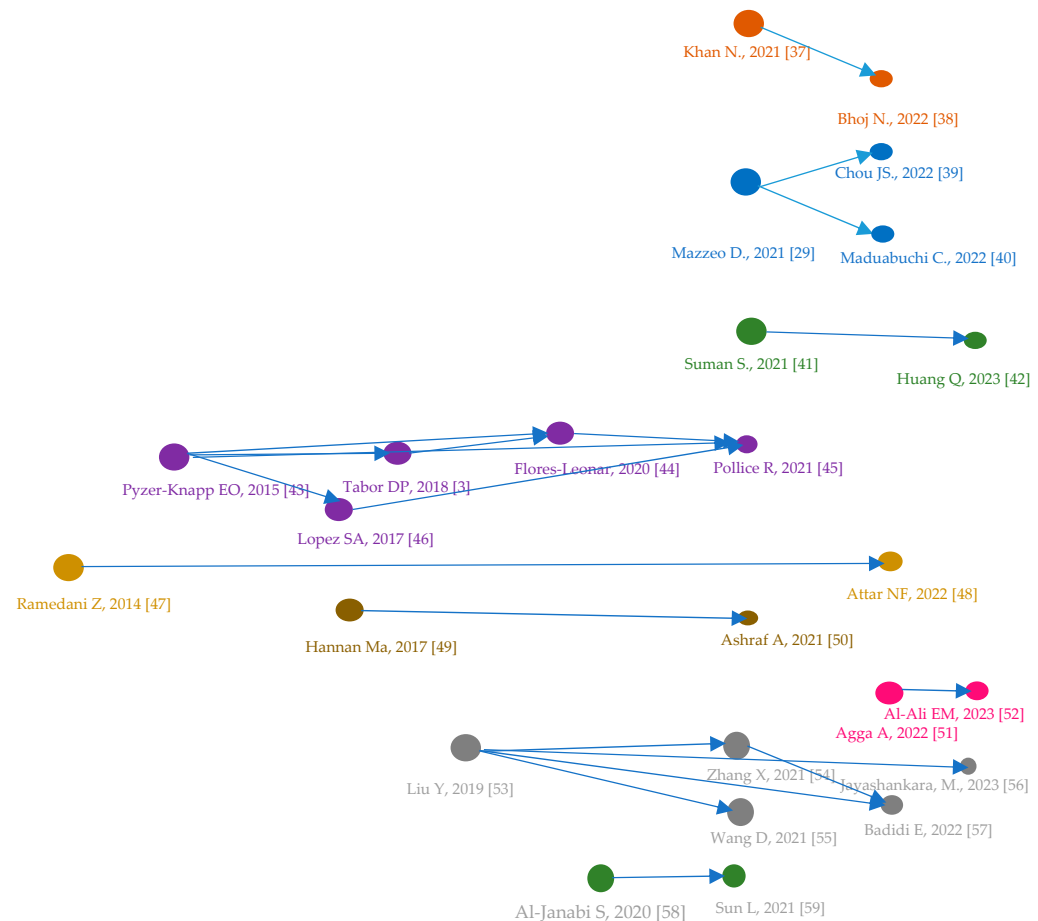


Figure 2. Global Citation Score (GCS) analysis of key articles on sustainable energy and artificial intelligence applications [3,29,37–59].

In Cluster 1, research harnesses AI to enhance the prediction of solar radiation with a particular focus on data from Iran, where machine learning techniques like Support Vector Regression and Fuzzy Linear Regression are pivotal, evidenced by Ramedani Z’s [47] influential 2014 work.

Cluster 2 spotlights the acceleration of clean energy materials discovery through AI, notably in organic photovoltaics and quantum chemistry. It emphasizes the significant strides made in automation and high-throughput methodologies, with Tabor et al. [3]’s paper being a cornerstone in the field.

The discussions in Cluster 3 revolve around the optimization of battery management systems, with innovative control algorithms for charge equalization in lithium-ion batteries, highlighting sustainable energy storage solutions and the extension of battery life.

Cluster 4 explores the synergy of IoT with AI in advancing energy management within smart urban environments, where edge computing and deep learning are critical for efficient energy utilization and predictive analytics in settings ranging from smart homes to city-wide systems.

Green energy generation and management form the crux of Cluster 5, delving into AI-driven wind energy optimization. The efficacy of convolutional neural networks in predicting energy output is explored, along with AI’s broader potential in diverse energy sectors.

Energy consumption prediction and management within buildings are the focal points of Cluster 6, where hybrid AI models, especially those combining CNNs with LSTM

networks, offer promising advancements in multi-step forecasting and the integration of photovoltaic power systems.

Cluster 7 provides a vista into the broad spectrum of AI applications within the clean energy and nuclear sectors, from community-level clean energy initiatives to complex nuclear reactor modelling, underscoring AI's role in optimization and predictive analysis.

AI's foray into nuclear reactor applications is scrutinized in Cluster 8, where papers discuss not only the current landscape but also the forward-looking potential of AI to revolutionize this domain, particularly through causal learning techniques.

Finally, Cluster 9 hones in on solar energy production forecasting. Here, the interplay of advanced deep learning architectures such as CNNs, LSTMs, and Transformers is proposed to significantly enhance the accuracy of production forecasts.

4.3. Co-Citation Network

A co-citation network is a type of network in which two nodes (in this case, papers) are connected if they are cited together by another paper. Table 4 provides information about specific nodes (papers) in this network, their assigned cluster, and three centrality metrics: betweenness, closeness, and PageRank. Betweenness centrality measures the extent to which a node acts as a bridge along the shortest paths between other nodes. A higher betweenness centrality indicates that the node is essential in connecting different parts of the network. Closeness centrality measures how close a node is to all other nodes in the network. A higher closeness centrality indicates that a node can reach other nodes in the network in fewer steps, suggesting its proximity to other nodes. The clusters were determined by using the walktrap algorithm. The walktrap algorithm is a method for community detection in networks. It is based on the idea that if you perform random walks on the graph, the walks are more likely to stay within the same community because there are only a few edges that lead outside a given community.

Table 4. Key authors and their influence in the network.

Node	Cluster	Betweenness	Closeness	PageRank
Hochreiter S. 1997 [60]	1	39	0.02	0.13
Lecun Y. 2015 [61]	1	0	0.01	0.05
Gers F.A. 1999 [62]	1	0	0.01	0.03
Liu H. 2018 [63]	1	0	0.01	0.02
Wen L. 2019 [64]	1	0	0.01	0.02
Breiman L. 2001 [65]	2	17	0.02	0.07
Hastie T. 2009 [66]	2	0	0.01	0.03
Lahouar A. 2017 [67]	2	0	0.01	0.04
Aburto L. 2007 [68]	3	0	0.01	0.05
Graves A. 2005 [69]	3	16	0.02	0.05

Figure 3 depicts the key authors and their influence in the network. Cluster 1 revolves around deep learning, given the seminal works of Hochreiter (LSTMs) [60], Lecun (CNNs) [61], and others. Hochreiter's 1997 paper on LSTMs has the highest betweenness centrality, indicating its role as a bridge in the network. This suggests that this paper connects various subfields or topics within this cluster. Cluster 2 seems to be centered around machine learning and statistical learning techniques, given the presence of Breiman's [65] and Hastie's [66] works. Breiman's 2001 [65] paper has a notable betweenness centrality, pointing to its influential role in bridging various topics or papers within this cluster.

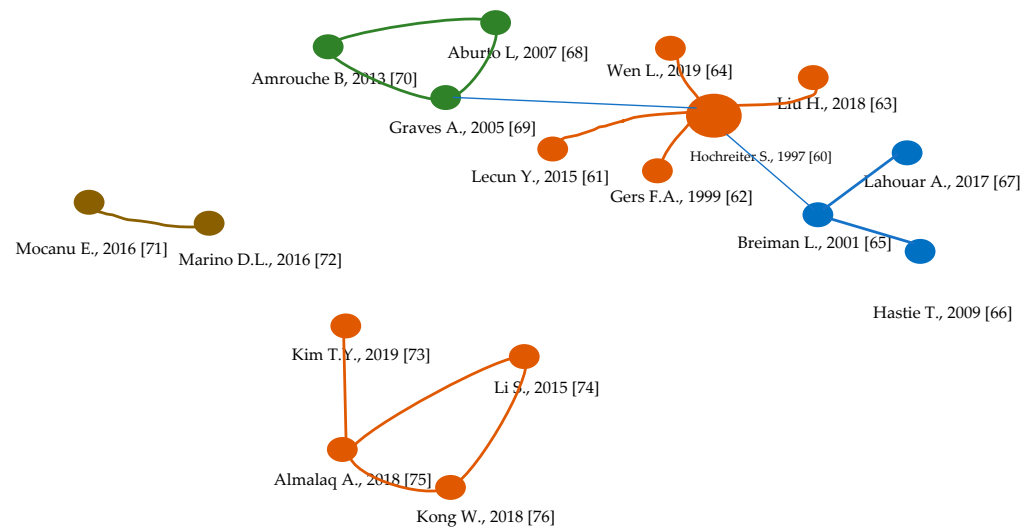


Figure 3. Key authors and their influence in the network [60–76].

The specific theme of Cluster 3 is not immediately clear from the provided authors and years. However, Graves’ 2005 [69] paper has a significant betweenness centrality, indicating its central role in this cluster.

In Cluster 4, the conversation extends to the innovation of green energy optimization, particularly highlighted by Almalag’s 2018 [75] paper, which not only contributes to the domain but also emerges as a pivotal piece, connecting various strands of research within the cluster as evidenced by its notable betweenness centrality.

4.4. Thematic Analysis

The results of the thematic analysis reveal insightful patterns and themes that shed light on the potential of artificial intelligence in advancing clean energy technologies in Europe. Table 5 presents various metrics for different clusters of research topics. The clusters are labelled based on their most representative keywords, such as “fuzzy inference”, “photovoltaic cells”, “artificial intelligence”, and so on. The metrics include Callon Centrality, Callon Density, Rank Centrality, Rank Density, and Cluster Frequency.

Table 5. Clusters resulted from thematic analysis.

Cluster	Callon Centrality	Callon Density	Rank Centrality	Rank Density	Cluster Frequency
Fuzzy inference	1.31	64.52	4	9	28
Photovoltaic cells	6.41	44.69	6	2	82
Artificial intelligence	34.42	41.84	10	1	679
Big data	9.45	46.89	7	3	144
Wind power	18.76	78.41	9	10	405
Deep learning	16.23	50.10	8	5	323
Controllers	1.47	60.51	5	8	30
Nanogenerators	0.70	57.63	2	6	28
Reinforcement learning	1.28	59.91	3	7	33
Automobiles	0	50	1	4	2

The establishment of clusters was based on a combination of keyword co-occurrence and thematic similarity. Using the walktrap algorithm, we analyzed the frequency with which certain keywords appeared together within the same articles. This algorithm identifies communities (or clusters) in networks by simulating random walks. In the context of our research, a “walk” represents a sequence of related topics or keywords. The walktrap algorithm identifies clusters by looking for densely connected groups of nodes, which in our case are keywords or topics.

The Callon Centrality metric gauges the importance of a cluster within the entire network. Clusters with higher Callon Centrality values are more central and influential. For instance, the “artificial intelligence” cluster has a high centrality value, indicating its pivotal role in the research landscape.

The Callon Density metric assesses the internal cohesion of a cluster. A higher Callon Density value suggests that the research topics within that cluster are closely related. For example, the “wind power” cluster has a high density, implying that the articles within this cluster are tightly interconnected in terms of their thematic content.

The Rank Centrality and Rank Density metrics provide a comparative ranking of clusters based on their centrality and density values, respectively. Lower rank values denote higher importance or cohesion.

The Cluster Frequency metric denotes the prevalence of research topics within a cluster. A higher frequency indicates that the cluster’s themes are more commonly discussed in the literature.

From our analysis, we observed that clusters with high centrality and density values, such as “artificial intelligence” and “wind power”, represent core areas of research that have consistently garnered attention over the years. On the other hand, clusters with lower frequency values, like “Automobiles”, might represent niche or emerging areas of research that have not yet reached mainstream prominence.

Furthermore, the interplay between centrality and density can offer insights into the nature of the research landscape. For instance, a cluster with a high centrality but low density might suggest a broad research area with diverse subtopics, while a cluster with high values for both metrics indicates a well-defined, influential research domain with closely related topics.

Figure 4 presents the clusters resulting from the thematic analysis of the collected data. Each cluster is identified by a set of keywords that characterize it. Figure 4 is useful in identifying and understanding the most important themes and research directions in the field of artificial intelligence and clean energy in Europe.

Our systematic review revealed several distinct yet interrelated research clusters. The Fuzzy Inference cluster intertwines fuzzy logic and neural networks with the sustainability potential of alternative energy sources like biogas and biomass. Research here delves into sophisticated control mechanisms and decision-making frameworks essential for optimizing these energy systems. Photovoltaic Cells form a vibrant cluster that pivots around harnessing solar energy through advanced photovoltaic technologies. This extends to innovative battery charging solutions for electric vehicles, reflecting a synergistic approach to solar power utilization. The Artificial Intelligence cluster spans a wide array of AI implementations, bridging renewable energy techniques with policy and sustainable development goals. It is a testament to AI’s overarching impact on the clean energy landscape. In the Big Data cluster, the focus sharpens on analytics and data processing methodologies that propel renewable energy and environmental protection into the digital age, highlighting the intersection between large-scale data handling and clean energy advancement.

Wind Power cluster encapsulates the dynamics of wind energy exploration, where forecasting and power generation methods, bolstered by machine learning and neural networks, are at the forefront, pushing the boundaries of predictive analytics in wind energy.

Deep Learning emerges as a pivotal cluster, spotlighting the deployment of sophisticated algorithms in enhancing energy efficiency across smart power grids and IoT landscapes, marking a new era of intelligent energy management.

The Controllers cluster converges on innovative energy management, delving into the control systems governing battery life and power conversion efficiency, underscoring the critical role of precision in sustainable energy systems.

Nanogenerators represent a frontier cluster that connects nanotechnology with sustainable energy production, reflecting the cross-disciplinary innovation needed to harness energy at the microscopic scale for agricultural and other sustainable applications.

Reinforcement Learning is distinguished by its focus on employing advanced algorithms for green computing, highlighting how intelligent resource allocation can lead to substantial gains in energy optimization.

Lastly, the Automobiles cluster concentrates on the intersection of energy consumption and sustainable practices within the automotive sector, steering the conversation towards greener transportation solutions.

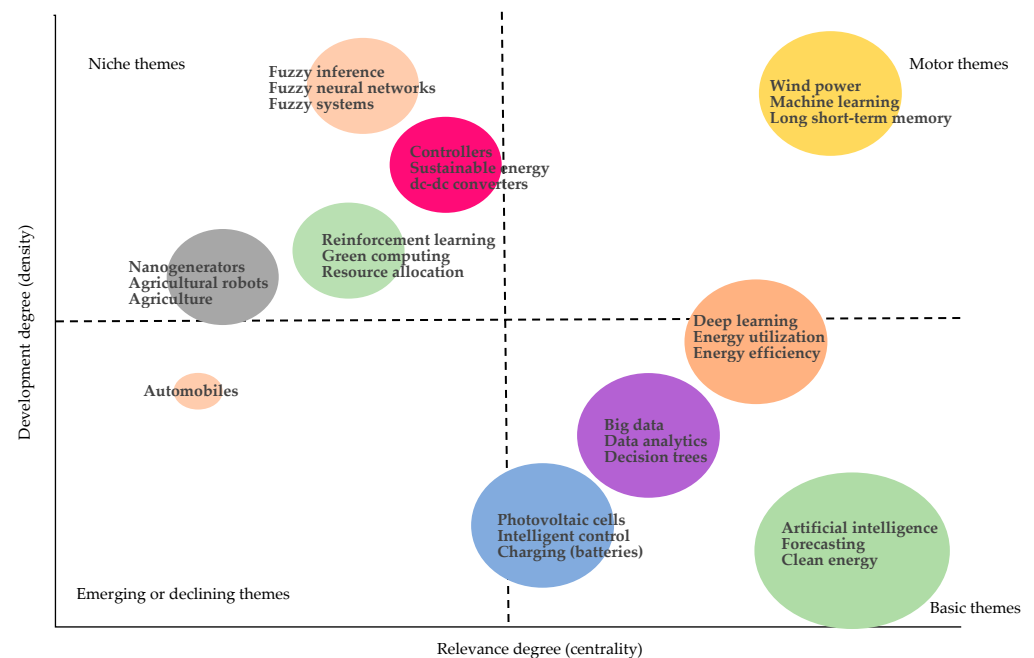


Figure 4. Clusters resulted from thematic analysis.

4.5. Thematic Evolution from 2006–2021 to 2022–2023

A selection of research topic connections between the 2006–2021 and 2022–2023 periods is depicted in Table 6. Each row represents a link between a research topic from the first period and another research topic from the second period, with a focus on the Weighted Inclusion Index and Stability Index.

In order to understand the table, we mention:

- **From and To:** These columns represent the thematic transition. The “From” column lists the dominant research topics from 2006 to 2021, while the “To” column indicates the corresponding or evolved topics in 2022–2023.
- **Words:** This column provides specific keywords associated with the research topics, offering a glimpse into the core focus of each theme.
- **Weighted Inclusion Index:** This metric quantifies the frequency of keywords from the “From” period appearing in the “To” period, adjusted for the significance of each keyword. A higher value suggests that the topic from the earlier period has a pronounced presence or influence in the latter period.
- **Stability Index:** This metric evaluates the consistency of the relationship between two research topics across both periods. A higher value indicates a more stable and enduring connection between the topics.

Table 6. Thematic evolution sorted by stability index (first 14 rows).

From	To	Words	Weighted Inclusion Index	Stability Index
global warming— 2006–2021	artificial neural network 2022–2023	global warming	1.00	0.17
agricultural robots— 2006–2021	nanogenerators 2022–2023	nanogenerators	0.25	0.08
machine learning— 2006–2021	artificial neural network 2022–2023	article	0.20	0.04
machine learning techniques— 2006–2021	data analytics 2022–2023	decision trees	0.13	0.04
machine learning techniques— 2006–2021	neural networks 2022–2023	fuzzy inference; fuzzy neural networks	0.17	0.04
sustainable development— 2006–2021	biofuels 2022–2023	biofuels	0.50	0.04
machine learning— 2006–2021	clean energy 2022–2023	economics	0.04	0.03
machine learning— 2006–2021	data analytics 2022–2023	human	0.05	0.03
photovoltaic cells— 2006–2021	artificial neural network 2022–2023	artificial neural network; photovoltaic system	0.40	0.03
sustainable development— 2006–2021	clean energy 2022–2023	sustainable development; environmental technology	0.25	0.03
sustainable development— 2006–2021	data analytics 2022–2023	data analytics	0.10	0.03
sustainable development— 2006–2021	Neural networks 2022–2023	greenhouse gases	0.06	0.03
artificial intelligence— 2006–2021	artificial neural network 2022–2023	performance assessment	0.13	0.02
artificial intelligence— 2006–2021	clean energy 2022–2023	clean energy; investments; planning	0.40	0.02

For instance, the first row indicates that the topic “global warming” from 2006 to 2021 has evolved or is closely related to “artificial neural network” in 2022–2023. The Weighted Inclusion Index of 1.00 suggests a strong connection, while the Stability Index of 0.17 indicates a moderate level of consistency in this thematic transition.

We enunciate the key insights from Table 4:

- Evolution of Topics: Some topics have undergone significant evolution. For instance, “global warming” has transitioned to studies involving “artificial neural networks”, suggesting a shift towards leveraging AI techniques to address climate change.
- Interdisciplinary Nature: Topics like “machine learning” from 2006 to 2021 show connections to diverse areas in 2022–2023, highlighting the interdisciplinary nature of research in this domain.
- Emergence of New Areas: Connections, such as that between “agricultural robots” and “nanogenerators”, hint at emerging research areas, potentially pointing to innovations at the intersection of agriculture and nanotechnology.

- **Stability of Themes:** While some connections remain stable over time, others might be transient, reflecting the dynamic nature of research in artificial intelligence and clean energy.

Following the thematic analysis, each research cluster was meticulously examined to gain a deeper understanding of the prevailing trends, methodologies, and applications within the realm of artificial intelligence and clean energy technologies in Europe. This granular analysis of individual clusters allowed for the identification of specific AI techniques, their applications, and the overarching themes that dominate the current research landscape.

4.6. Findings on Advancements in AI Techniques for Renewable Energy

Overall, the findings highlight the dynamic interplay among research topics within the renewable energy domain over time, revealing the progression of certain areas, the convergence of others, and the emergence of novel research trends.

The application of machine learning and artificial intelligence in predicting renewable energy outputs has made remarkable strides. For instance, Ramedani et al. (2014) [47] demonstrate the utility of radial basis function-based support vector regression in predicting global solar radiation, while Zhang et al. (2021) [54] illustrate the integration of IoT systems for enhancing green energy in smart cities. These advancements underscore how significantly the accuracy of renewable energy prediction has improved.

Moreover, deep learning and neural networks are at the forefront of refining short-term energy forecasts and enhancing energy management. Alhussein et al. (2019) [77] discuss a microgrid-level energy management approach that capitalizes on short-term forecasting, and Han (2021) [78] presents an efficient framework for intelligent energy management in IoT networks. Putz et al. (2021) [79] also contribute to this discussion with their novel approach to wind power forecasting using deep neural architectures.

In the broader context of the renewable energy sector, AI is positioned as a transformative force. Serban and Lytras (2020) [80] explore AI's role in developing smart renewable energy infrastructures for Europe's next-generation smart cities. Paiho (2021) [81] investigates the potential of cross-commodity energy-sharing communities, offering insights into the market, regulatory, and technical landscapes. These studies collectively affirm the potential of AI to revolutionize renewable energy systems, aligning with the sector's innovative trajectory.

The integration of machine learning techniques in energy management systems can contribute to minimizing losses, optimizing cost, and improving energy efficiency [82,83].

Advanced analytics and data-driven approaches can facilitate the discovery of new materials for clean energy applications [3].

Deep learning-based systems have been applied to monitor and maintain renewable energy infrastructures, such as solar photovoltaic farms and wind energy plants [84].

In recent years, there has been a growing interest in green energy, sustainable technologies, and artificial intelligence (AI) for optimizing energy systems. For instance, Geetha et al. (2022) [85] proposed a green energy-aware and cluster-based communication system for future load prediction in the Internet of Things (IoT). This system aims to improve energy efficiency and reduce environmental impacts.

Digitalization of the energy sector is another key trend, as highlighted by Singh et al. (2022) [86] in their research on Energy System 4.0. The authors discussed how digital technologies can be leveraged to enhance sustainability in the energy sector. In the context of smart cities, Badidi (2022) [57] examined the promise and potential of edge AI and blockchain for developing smart, sustainable urban environments.

AI and machine learning (ML) techniques have been employed to optimize energy management and forecasting in various contexts. Ye et al. (2022) [87] developed Dynamic-Net, a time-variant ODE network for multi-step wind speed prediction, showcasing the potential of deep learning for accurate forecasting in the renewable energy domain.

The integration of artificial intelligence (AI) techniques into renewable energy systems has been a significant area of research in recent years. These techniques offer promising solutions to optimize energy production, consumption, and storage. Table 7 provides a comprehensive overview of the recent advancements in the application of AI techniques in renewable energy and related fields. The studies referenced highlight the potential of AI in enhancing the efficiency, sustainability, and innovation in renewable energy systems and material discovery.

Table 7. Overview of AI techniques in renewable energy and related fields.

Technique	Application	Reference	Key Findings
Neural Networks	Wind Energy Prediction	Putz et al. (2021) [79]	Enhanced efficiency and reduced maintenance costs of wind farms through deep neural architecture.
Deep Learning (CNN)	Solar Energy Forecasting	Ramedani et al. (2014) [47]	Accurate prediction of solar irradiance using support vector regression and various data sources.
Deep Learning (CNN)	Energy Storage Optimization	Yuce et al. (2016) [82]	Optimized energy management in the domestic sector through ANN-GA smart appliance scheduling.
Deep Learning (CNN)	Energy Consumption Analysis	Tabor et al. (2018) [3]	Accelerated discovery of materials for clean energy through smart automation and AI analysis.
Neural Networks	Organic Photovoltaics Design	Tabor et al. (2018) [3]	Identification of top non-fullerene acceptor candidates.
Neural Networks	Materials Discovery	Pyzer-Knapp EO et al. (2015) [43]	Acceleration of materials discovery using insights from the Harvard Clean Energy Project.

This synthesis provides an overview of the recent advancements in the application of AI techniques in renewable energy. The referenced studies highlight the potential of AI in enhancing the efficiency and sustainability of renewable energy systems.

AI Techniques in Clean Energy and Renewable Energies

The integration of artificial intelligence (AI) and other advanced techniques in the realm of clean energy and renewable energies is evident from recent research. Table 8 presents a breakdown of the studies:

This table provides a structured overview of the recent advancements in the application of AI and other techniques in the fields of clean energy and renewable energies. Each entry highlights the core focus of the respective study and its key findings.

In the realm of artificial intelligence, it is easy to become encumbered by the broad terminologies of “machine learning” and “deep learning”. However, upon closer inspection, we uncover a myriad of specific algorithms that have carved their niches within the expansive world of AI. Particularly in the domain of clean energy, some techniques have consistently proven to be more effective, adaptive, and promising than others. In this section, we will unravel the layers of abstraction and delve into the specific AI techniques that have been instrumental in driving innovations in clean energy.

- Convolutional Neural Networks (CNNs): Originally designed for image recognition tasks, CNNs have found relevance in clean energy by aiding in the optimization of solar panel placements, identifying patterns in energy consumption, and predicting potential system failures based on visual cues.
- Recurrent Neural Networks (RNNs): With their ability to remember previous data in a sequence, RNNs have become a cornerstone for predicting energy consumption based on historical patterns, especially in grid management and demand forecasting.

- Generative Adversarial Networks (GANs): Although traditionally used for generating data, GANs have been employed in clean energy for simulating various energy scenarios, aiding researchers in understanding potential outcomes without the need for real-world testing.

Table 8. AI techniques in clean energy and renewable energies.

Technique	Application	Reference	Key Findings
Clean Energy	Materials Acceleration	Flores-Leonar MM et al. (2020) [44]	Development of platforms for autonomous experimentation.
Clean Energy	Renewable Energy in Africa	Amir M, Khan SZ (2021) [88]	Assessment of renewable energy's status and challenges in Africa.
Clean Energy	Molecular Doping Efficiency	Yan H, Ma W (2022) [89]	Fundamental principles and strategies for molecular doping in organic semiconductors.
Clean Energy	Sustainable Development Goals	Ebolor A et al. (2022) [90]	Technologies underpinned by frugal innovation to foster sustainable development goals.
Clean Energy	Aerospace Engineering and SDGs	Sánchez-Roncero A et al. (2023) [91]	Critical analysis of the intersection between aerospace engineering and sustainable development goals through AI.
Clean Energy	Electric Power ESG Issues	Bai S, Zhang J (2022) [92]	Management and disclosure of electric power environmental and social governance issues in the AI era.
Clean Energy	Sustainable Clothing Design	Dai J et al. (2022) [93]	Designing energy color-changing clothing with a focus on environmental protection and sustainable development.
Renewable Energies	Local Neighborhood Energy Planning	Hettinga S et al. (2018) [94]	A decision support system for energy planning in local neighborhoods.
Renewable Energies	Digitalization of Energy Sector	Singh R et al. (2022) [86]	Exploration of the digital transformation of the energy sector with a focus on sustainability.
Renewable Energies	Digital Twin Technologies in Energy	Ghenai C et al. (2022) [95]	Comprehensive review of recent trends in digital twin technologies in the energy sector.

To facilitate a better understanding, Table 9 provides a detailed breakdown of these AI techniques, their core functionalities, applications in clean energy, and key reference studies.

This enhanced focus on specific AI techniques accentuates the versatility and adaptability of AI in addressing the multifaceted challenges presented by the clean energy sector.

In summary, the convergence of green energy initiatives, sustainable technology deployment, and AI-driven optimization of energy systems is becoming increasingly prominent in the recent literature. This body of work underscores the pivotal role of digitalization in energy sectors, particularly within the realms of smart cities and advanced energy management systems.

Geetha (2022) [85] provides insights into green energy-aware communication strategies crucial for future load prediction in IoT environments, a fundamental component for smart city infrastructure. Singh (2022) [86] discusses the broader concept of Energy System 4.0, emphasizing the digital transformation of the energy sector with a particular focus on sustainability, which aligns with the digitalization narrative.

Badidi (2022) [57] explores the integration of Edge AI and Blockchain technology in smart sustainable cities, underscoring the promise and potential of these technologies in

enhancing urban energy systems. Ye (2022) [87] contributes to the field of energy forecasting by introducing DynamicNet, a novel time-variant ODE network aimed at improving the accuracy of multi-step wind speed predictions.

These studies collectively illuminate the advancing trend towards integrating AI into the core of energy system optimization, establishing a strong foundation for future innovations that prioritize efficiency and sustainability.

Table 9. AI techniques in clean energy.

AI Technique	Description	Application in Clean Energy	Key References
CNNs (Convolutional Neural Networks)	Specialized in handling grid-like data structures such as images. Employs spatial hierarchies to detect patterns.	Optimal solar panel placement, energy consumption pattern recognition, system failure predictions.	Ramedani, Z. et al. (2014) [47], Zhang, X., Manogaran, G., and Muthu, B. (2021) [54]
RNNs (Recurrent Neural Networks)	Designed to handle sequential data, possessing the ability to remember past data points.	Energy consumption forecasting, grid management.	Alhussein, M., Haider, S.I., and Aurangzeb, K. (2019) [40], Han, T. et al. (2021) [77]
GANs (Generative Adversarial Networks)	Comprises two networks, one generating data while the other discerns its authenticity.	Simulating energy scenarios, generating potential clean energy system designs.	Pyzer-Knapp, E.O., Li, K., and Aspuru-Guzik, A. (2015) [43], Li, C. (2021) [33]

5. Conclusions

In conclusion, this study has provided valuable insights into the evolving landscape of research topics related to renewable energy, artificial intelligence, and sustainable technologies. By analyzing clusters of research topics and their connections across different time periods, we have identified key trends, interdisciplinary research, emerging areas, and the evolution of research focus in the field.

The research landscape from 2006 to 2023 reveals a robust and active research community in Europe, focusing on the application of AI in clean energy. The high volume of articles compared to reviews suggests a significant amount of primary research being conducted. Furthermore, the international co-authorship percentage underscores the collaborative nature of this research, emphasizing global interest in AI and clean energy.

The predefined categories derived from frequently encountered terms in the literature provide a structured approach to understanding the primary areas of focus in AI and clean energy. These categories range from specific AI techniques to different energy types, management strategies, and broader societal implications.

In the temporal trends of key topics in energy and AI, there is a clear shift towards sustainable energy solutions and the integration of advanced AI technologies. Topics like artificial intelligence, deep learning, and machine learning have seen a significant surge in recent years, indicating the growing integration of these technologies in the energy sector.

The historiography analysis, using the Global Citation Score (GCS), provides insights into the impact and recognition of specific articles within the broader academic community. The clusters identified, such as Solar Radiation Prediction using AI and AI in Material Discovery for Clean Energy, highlight the diverse applications of AI in the energy sector.

The co-citation network analysis reveals the interconnectedness of research papers and their influence in the domain of AI and clean energy. The centrality metrics, including betweenness, closeness, and PageRank, provide insights into the influence and connectivity of specific nodes (papers) within the network. For instance, Cluster 1, centered around deep learning, showcases the seminal works of Hochreiter and Lecun, with Hochreiter's 1997 paper on LSTMs playing a pivotal role in connecting various subfields within this cluster.

In summary, the research landscape from 2006 to 2023 in Europe, focusing on AI's application in clean energy, is vibrant and collaborative. The thematic and trend analyses

underscore the growing importance of AI techniques in the energy sector, with a clear shift towards sustainable solutions. The historiography and co-citation network analyses provide a deeper understanding of the influential works and their interconnectedness in this domain. As the field continues to evolve, these insights offer a foundation for future research endeavors and collaborations in the intersection of AI and clean energy.

The thematic analysis results offer a comprehensive understanding of the role of artificial intelligence in propelling clean energy technologies in Europe. Table 5 showcases metrics for distinct clusters of research topics, labelled by representative keywords like “fuzzy inference”, “photovoltaic cells”, and “artificial intelligence”.

Clusters like “artificial intelligence” and “wind power” with high centrality and density values represent core research areas. Conversely, lower frequency clusters, such as “automobiles”, might signify niche or emerging research areas.

The findings elucidate the relationships between research topics across different periods, revealing how research areas have evolved or merged. AI’s application in predicting renewable energy production has seen significant improvement. Advanced algorithms, including deep learning and neural networks, have enhanced short-term forecasts and energy management. AI holds the potential to transform the renewable energy sector, with applications in smart energy infrastructures and energy-sharing communities. The integration of machine learning in energy management systems can optimize cost and improve energy efficiency. Advanced analytics can aid in discovering new materials for clean energy applications.

Notably, machine learning and artificial intelligence have emerged as essential tools for predicting renewable energy production, optimizing energy management, and enhancing energy efficiency in various applications. The advancements in deep learning and neural networks have demonstrated promising results in improving short-term forecasts, energy management in smart cities, and IoT networks. Furthermore, AI has shown its potential to revolutionize the renewable energy sector, with applications in smart energy infrastructures and cross-commodity energy-sharing communities.

The integration of machine learning techniques in energy management systems has proven beneficial in minimizing losses, optimizing cost, and improving energy efficiency. Advanced analytics and data-driven approaches have facilitated the discovery of new materials for clean energy applications, while deep learning-based systems have been applied to monitor and maintain renewable energy infrastructures, such as solar photovoltaic farms and wind energy plants.

Recent research has also highlighted the increasing importance of green energy, sustainable technologies, and AI in optimizing energy systems, with a focus on digitalization, smart cities, and improved energy management and forecasting. The growing interest in these areas indicates that future research will likely continue to explore novel applications, interdisciplinary connections, and innovative solutions to address the challenges of sustainability, energy efficiency, and environmental protection.

5.1. Research Limitations

The research primarily focused on studies and data available up to 2023. Subsequent developments and studies post this period were not considered. While the research aimed to provide insights into Europe’s clean energy transition, specific regional nuances within Europe might not have been deeply explored. The rapid evolution of AI and clean energy technologies means that newer advancements might not be covered in this research. Interpretations of certain data points and studies might carry inherent biases or subjectivity, despite efforts to maintain objectivity.

5.2. Future Research Directions and Prospects

The dynamic nature of technology, especially at the nexus of artificial intelligence and clean energy, offers a plethora of avenues for further exploration. As Europe continues its journey towards sustainable energy solutions, a more granular examination of individual

regions or countries within its boundaries could yield nuanced insights. Conversely, broadening the scope to encapsulate global perspectives might reveal universal strategies or region-specific challenges in harnessing AI for clean energy.

Continuous advancements in both AI and clean energy necessitate regular updates in research methodologies and findings. Collaborative endeavors with industry experts can bridge the gap between theoretical research and practical application, shedding light on tangible challenges and advantages of AI integration in clean energy systems. To enrich the depth and breadth of future studies, researchers might consider integrating diverse data sources such as industry analytics, expert testimonials, and detailed case studies.

Emerging technologies also present intriguing prospects for the AI–clean energy interface. For instance, the amalgamation of quantum computing with AI could revolutionize the field. Given quantum computers’ unique computational capabilities, they could significantly accelerate AI-driven simulations, optimizations, and even discoveries of novel energy-efficient materials.

Neuromorphic computing, inspired by human neural structures, offers a promising avenue for energy-efficient AI operations. Implementing such systems in clean energy solutions might facilitate real-time monitoring and decision-making processes, sidestepping the substantial energy expenditures typical of conventional AI algorithms.

Furthermore, the advent of edge computing, which emphasizes processing data closer to its source, promises reduced latency. This is paramount for instantaneous energy monitoring and swift adaptive measures, a feature especially vital for efficient smart grid operations.

By steering research towards these promising horizons, we can aspire for an AI–clean energy synergy that is not only at the forefront of technological innovation but also champions sustainability and meaningful impact.

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